Seminar on Positivity in Algebraic Geometry

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1 Inequalities

1.1 Global Results

Theorem 1.1 (Generalized inequality of Hodge type)**.** Let X be an irreducible complete variety (or scheme) of dimension n, and let

$$\delta_1, \cdots, \delta_n \in \mathrm{Nef}(X)$$

be nef classes. Then

$$(\delta_1 \cdots \delta_n)^n \geq ((\delta_1)^n) \cdots ((\delta_n)^n)\,. \tag{1}$$

Proof. We can assume first of all that X is reduced since in general its cycle satisfies $[X] = a \cdot [X_{\mathrm{red}}]$ for some $a > 0$ [Ful98, §1.5]. By Chow's Lemma [Har77, II, Exercise 4.10] and the projection formula we can also suppose X is projective. In this case, it suffices to prove the theorem under the additional assumption that the δ_i are ample. In fact, if the stated inequality holds for $\delta_i \in \mathrm{Amp}(X)$, then by continuity it holds also for $\delta_i \in \mathrm{Nef}(X) = \overline{\mathrm{Amp}(X)}$.

We now argue by induction on $n = \dim X$. If X is a smooth projective surface, then the stated inequality is a version of the Hodge Index Theorem (cf. [Har77, V, Exercise 1.9]). As in the previous paragraph, once one knows (1) for ample classes, it follows also for nef ones. This being said, if X is singular, one deduces (1) by passing to a resolution. We assume henceforth that $n \geq 3$, and that (1) is already known on all irreducible varieties of dimension $n-1$.

We next claim that given any ample classes

$$\beta_1, \cdots, \beta_{n-1}, h \in \mathrm{Amp}(X)$$

one has the inequality

$$(\beta_1 \cdots \beta_{n-1} \cdot h)^{n-1} \geq \big((\beta_1)^{n-1} \cdot h\big) \cdots \big((\beta_{n-1})^{n-1} \cdot h\big)\,. \tag{2}$$

In fact, it suffices by continuity to prove this when h and the β_i are rational ample classes. By multiplying a sufficiently large constant, we may assume that they are represented by very ample divisors $B_1, \cdots, B_{n-1}$ and H on X, and the issue is to verify the inequality

$$(B_1 \cdots B_{n-1} \cdot H)^{n-1} \geq \left((B_1)^{n-1} \cdot H\right) \cdots \left((B_{n-1})^{n-1} \cdot H\right).$$

We suppose that H is an irreducible projective scheme of dimension $n-1$, and that each B_i meets H properly. Denoting by $\overline{B_i}$ the restriction of B_i to H, the inequality in question is equivalent term by term to the relation

$$\left(\overline{B_1} \cdots \overline{B_{n-1}}\right)^{n-1} \geq \left((\overline{B_1})^{n-1}\right) \cdots \left((\overline{B_{n-1}})^{n-1}\right)$$

of intersection numbers on H. But this follows by applying the induction hypothesis to H.

We now show that the desired inequality (1) follows formally from (2). So let

$$\delta_1, \cdots, \delta_n \in \mathrm{Amp}(X)$$

be n ample classes on X (The amplitude of the δ_i guarantees that all of the intersection numbers appearing in the computations that follow are positive, see Example 1.2.5. Therefore we can cancel and take roots without further thought). Fix some index $j \in \{1, \cdots, n\}$ and apply (2) with $h = \delta_j$ and $\beta_1, \cdots, \beta_{n-1}$ the remaining δ_i. One finds

$$(\delta_1 \cdots \delta_n)^{n-1} \geq \prod_{i \neq j} \left(\delta_i^{n-1} \cdot \delta_j\right).$$

Taking the product over j yields

$$(\delta_1 \cdots \delta_n)^{n(n-1)} \geq \prod_j \prod_{i \neq j} \left(\delta_i^{n-1} \cdot \delta_j\right). \tag{3}$$

But now apply (2) with $h = \beta_1 = \cdots = \beta_{n-2} = \delta_i$ and $\beta_{n-1} = \delta_j$ to obtain

$$\left(\delta_i^{n-1} \cdot \delta_j\right)^{n-1} \geq (\delta_i^n)^{n-2} \left(\delta_i \cdot \delta_j^{n-1}\right).$$

Therefore

$$\begin{aligned} \prod_j \prod_{i \neq j} \left(\delta_i^{n-1} \cdot \delta_j\right)^{n-1} &\geq \prod_j \prod_{i \neq j} (\delta_i^n)^{n-2} \left(\delta_i \cdot \delta_j^{n-1}\right) \\ &= \left(\prod_i (\delta_i^n)^{(n-1)(n-2)}\right) \left(\prod_j \prod_{i \neq j} \left(\delta_j \cdot \delta_i^{n-1}\right)\right). \end{aligned}$$

The second term on the right cancels against the left-hand side, and taking $(n-2)$nd roots one arrives at

$$\prod_j \prod_{i \neq j} (\delta_i^{n-1} \cdot \delta_j) \geq \prod_i (\delta_i^n)^{(n-1)}.$$

The inequality (1) follows by plugging this into (3) and taking $(n-1)$st roots. □

We record some variants and special cases. To begin with, one has the following:

Variant 1.2. Let X be an irreducible complete variety or scheme of dimension n, and fix an integer $0 \leq p \leq n$. Let

$$\alpha_1, \cdots, \alpha_p, \beta_1, \cdots, \beta_{n-p} \in \mathrm{Nef}(X)$$

be nef classes. Then

$$(\alpha_1 \cdots \alpha_p \cdot \beta_1 \cdots \beta_{n-p})^p \geq (\alpha_1^p \cdot \beta_1 \cdots \beta_{n-p}) \cdots \left(\alpha_p^p \cdot \beta_1 \cdots \beta_{n-p}\right). \tag{4}$$

Proof. Assume $\beta_1, \cdots, \beta_{n-p}$ are represented by very ample divisors $B_1, \cdots, B_{n-p}$ on X. Apply Theorem 1.1 on $B_1 \cap \cdots \cap B_{n-p}$ we obtain

$$(\overline{\alpha_1} \cdots \overline{\alpha_p})^p \geq ((\overline{\alpha_1})^p) \cdots ((\overline{\alpha_p})^p).$$

That is,

$$(\alpha_1 \cdots \alpha_p \cdot \beta_1 \cdots \beta_{n-p})^p \geq (\alpha_1^p \cdot \beta_1 \cdots \beta_{n-p}) \cdots \left(\alpha_p^p \cdot \beta_1 \cdots \beta_{n-p}\right).$$

□

The case of two classes is particularly useful:

Corollary 1.3 (Inequalities for two classes)**.** Let X be an irreducible complete variety or scheme of dimension n, and let $\alpha, \beta \in \mathrm{Nef}(X)$ be nef classes on X. Then the following inequalities are satisfied:

(i). For any integers $0 \leq q \leq p \leq n$,

$$\left(\alpha^q \cdot \beta^{n-q}\right)^p \geq \left(\alpha^p \cdot \beta^{n-p}\right)^q \cdot (\beta^n)^{p-q}. \tag{5}$$

(ii). For any $0 \leq i \leq n$,

$$\left(\alpha^i \cdot \beta^{n-i}\right)^n \geq (\alpha^n)^i \cdot (\beta^n)^{n-i}. \tag{6}$$

(iii).

$$((\alpha+\beta)^n)^{1/n} \geq ((\alpha^n))^{1/n} + ((\beta^n))^{1/n}. \tag{7}$$

Proof. For (i), take $\alpha_1 = \cdots = \alpha_q = \alpha$ and $\alpha_{q+1} = \cdots = \alpha_p = \beta_1 = \cdots = \beta_{n-p} = \beta$ in Variant 1.2.

Statement (ii) is the special case $q = i$ and $p = n$ of (i).

For (iii) expand out $(\alpha + \beta)^n$ and apply (ii), we have

$$\begin{aligned}((\alpha+\beta)^n) &= \sum_{i=0}^{n} \binom{n}{i} \left(\alpha^i \cdot \beta^{n-i}\right) \\ &\geq \sum_{i=0}^{n} \binom{n}{i} \left(\alpha^n\right)^{i/n} \cdot \left(\beta^n\right)^{(n-i)/n} \\ &= \left(\left((\alpha^n)\right)^{1/n} + \left((\beta^n)\right)^{1/n}\right)^n.\end{aligned}$$

Take nth roots get the assertion. □

Example 1.4 (Inequalities of Khovanskii and Teissier)**.** In the situation of Corollary 1.3, put $s_i = \left(\alpha^i \cdot \beta^{n-i}\right)$. Then for all $1 \leq i \leq n-1$,

$$s_i^2 \geq s_{i-1} \cdot s_{i+1}.$$

In other words, the function $i \mapsto \log s_i$ is concave.

Proof. Apply Variant 1.2 with $p = 2$, $\alpha_1 = \alpha$, $\alpha_2 = \beta$, $\beta_1 = \cdots = \beta_{i-1} = \alpha$, and $\beta_i = \cdots = \beta_{n-p} = \beta$. □

Remark 1.5 (Positive characteristics)**.** The material in this section remains valid for varieties defined over an algebraically closed field of arbitrary characteristic.

1.2 Mixed Multiplicities

Let X be a variety or scheme of pure dimension n, and let $x \in X$ be a (closed) point with maximal ideal $\mathfrak{m} \subset \mathcal{O}_X$. We suppose given $k \leq n$ ideal sheaves

$$\mathfrak{a}_1, \cdots, \mathfrak{a}_k \subset \mathcal{O}_X \text{ with } \operatorname{Supp}\left(\mathcal{O}_X/\mathfrak{a}_i\right) = \{x\},$$

i.e., we assume that the $\mathfrak{a}_i$ are $\mathfrak{m}$-primary. Fix also non-negative integers $d_1, \cdots, d_k$ with $\sum d_i = n$. Then one can define the **mixed multiplicity**

$$e\left(\mathfrak{a}_1^{[d_1]}; \mathfrak{a}_2^{[d_2]}; \cdots; \mathfrak{a}_k^{[d_k]}\right) \in \mathbb{N}$$

of the $\mathfrak{a}_i$, for instance by the property that the lengths

$$\operatorname{length}_{\mathcal{O}_X}\left(\mathcal{O}_X/\mathfrak{a}_1^{t_1} \cdots \mathfrak{a}_k^{t_k}\right)$$

are given for $t_i \gg 0$ by a polynomial of the form

$$\sum_{d_1+\cdots+d_k=n} \frac{e\left(\mathfrak{a}_1^{[d_1]};\cdots;\mathfrak{a}_k^{[d_k]}\right)}{d_1!\cdots d_k!} t_1^{d_1}\cdots t_k^{d_k} + (\text{lower degree terms})\,.$$

So for example $e\left(\mathfrak{a}^{[n]}\right) = e(\mathfrak{a})$ is the classic Samuel multiplicity of a (viewed as an ideal in the local ring $\mathcal{O} = \mathcal{O}_{X,x}$). When X is affine, and one takes d_i "general" elements $g_{i,1},\cdots,g_{i,d_i} \in \mathfrak{a}_i$, then $e\left(\mathfrak{a}_1^{[d_1]};\mathfrak{a}_2^{[d_2]};\cdots;\mathfrak{a}_k^{[d_k]}\right)$ computes the intersection multiplicity at x of the corresponding divisors. More geometrically, let

$$\mu\colon X' \longrightarrow X$$

be a proper birational map that dominates the blow-up of each $\mathfrak{a}_i$ (i.e., μ factors through every $\mathrm{Bl}_{\mathfrak{a}_i} X$). From [Har77, II, 7.13 (a)] $\mathfrak{a}_i\mathcal{O}_{\mathrm{Bl}_{\mathfrak{a}_i} X}$ is an invertible sheaf, so the pullback $\mathfrak{a}_i\mathcal{O}_{X'}$ is also an invertible sheaf, $\mathfrak{a}_i\mathcal{O}_{X'} = \mathcal{O}_{X'}(-f_i)$ for some effective Cartier divisor f_i on X' contracting to x. Then

$$e\left(\mathfrak{a}_1^{[d_1]};\mathfrak{a}_2^{[d_2]};\cdots;\mathfrak{a}_k^{[d_k]}\right) = (-1)\cdot\left((-f_1)^{d_1}\cdots(-f_k)^{d_k}\right).$$

Note that by allowing repetitions, it is enough to study the multiplicities $e\left(\mathfrak{a}_1;\cdots;\mathfrak{a}_n\right)$ associated to n (possibly non-distinct) ideals $\mathfrak{a}_1,\cdots,\mathfrak{a}_n \subset \mathcal{O}_X$.

Example 1.6 (Samuel multiplicity of a product)**.** Given $\mathfrak{m}$-primary ideals $\mathfrak{a},\mathfrak{b} \subset \mathcal{O}_X$ as above, the Samuel multiplicity $e(\mathfrak{a}\mathfrak{b})$ of their product has the expression

$$e(\mathfrak{a}\mathfrak{b}) = \sum_{i=0}^{n} \binom{n}{i} e\left(\mathfrak{a}^{[i]};\mathfrak{b}^{[n-i]}\right)$$

in terms of the mixed multiplicities of $\mathfrak{a}$ and $\mathfrak{b}$.

Theorem 1.7 (Inequalities for mixed multiplicities)**.** Fix $x \in X$ as above.

(i). For any $\mathfrak{m}$-primary ideals $\mathfrak{a}_1,\ldots,\mathfrak{a}_n \subseteq \mathcal{O}_X$, one has

$$e(\mathfrak{a}_1;\cdots;\mathfrak{a}_n)^n \leq e(\mathfrak{a}_1)\cdots e(\mathfrak{a}_n).$$

(ii). Given $\mathfrak{m}$-primary ideals $\mathfrak{a},\mathfrak{b} \subseteq \mathcal{O}_X$, and any integers $0 \leq q \leq p \leq n$,

$$e\left(\mathfrak{a}^{[q]};\mathfrak{b}^{[n-q]}\right)^p \leq e\left(\mathfrak{a}^{[p]};\mathfrak{b}^{[n-p]}\right)^q \cdot e(\mathfrak{b})^{p-q}.$$

(iii). With $\mathfrak{a},\mathfrak{b} \subseteq \mathcal{O}_X$ as in (ii), and $0 \leq i \leq n$,

$$e\left(\mathfrak{a}^{[i]};\mathfrak{b}^{[n-i]}\right)^n \leq e(\mathfrak{a})^i \cdot e(\mathfrak{b})^{n-i}.$$

(iv). In the situation of (iii), set $m_i = e(\mathfrak{a}^{[i]}; \mathfrak{b}^{[n-i]})$. Then for $1 \leq i \leq n-1$,

$$m_i^2 \leq m_{i-1} \cdot m_{i+1}.$$

Remark 1.8 (Reversal of inequalities)**.** Note that compared to the global statements, the direction of the inequalities here is reversed. This may be explained by noting that the global results finally come down to the fact that the intersection form has signature $(1,-1)$ on a two-dimensional space of ample classes on a surface. One can view the local inequalities, on the other hand, as ultimately springing from the negativity of the intersection form on the space spanned by the exceptional curves in a birational map of surfaces. Of course, the classical Hodge index theorem is at work in both cases.

Example 1.9 (An inequality of Teissier)**.** Given $\mathfrak{m}$-primary ideals $\mathfrak{a}, \mathfrak{b} \subset \mathcal{O}_X$ as in Theorem 1.7, the Samuel multiplicities of $\mathfrak{a}, \mathfrak{b}$, and $\mathfrak{ab}$ satisfy the inequality

$$e(\mathfrak{ab})^{1/n} \leq e(\mathfrak{a})^{1/n} + e(\mathfrak{b})^{1/n}.$$

Proof. Thanks to Theorem 1.7 (iii), we have

$$\begin{aligned} e(\mathfrak{ab}) &= \sum_{i=0}^{n} \binom{n}{i} e\left(\mathfrak{a}^{[i]}; \mathfrak{b}^{[n-i]}\right) \\ &\leq \sum_{i=0}^{n} \binom{n}{i} e(\mathfrak{a})^{i/n} \cdot e(\mathfrak{b})^{(n-i)/n} \\ &= \left(e(\mathfrak{a})^{1/n} + e(\mathfrak{b})^{1/n}\right)^n. \end{aligned}$$

Take nth roots get the assertion. □

Example 1.10. Consider polynomials $f_1, \cdots, f_n \in \mathbb{C}[z_1, \cdots, z_n]$ defining a finite map $f\colon \mathbb{C}^n \to \mathbb{C}^n$, and let $g_1, \cdots, g_{n-1}$ be general elements in the ideal generated by the f_i. Assume that $f_1, \cdots, f_n$ all vanish at the origin, so that $f(0) = 0$. The g_i cut out a curve $\Gamma \subset \mathbb{C}^n$ which is typically singular at 0. But the multiplicity of Γ at 0 is bounded by the degree of f:

$$\operatorname{mult}_0(\Gamma) \leq (\deg f)^{1-\frac{1}{n}}.$$

Proof. Take $\mathfrak{a} = (z_1, \cdots, z_n)$, $\mathfrak{b} = (f_1, \cdots, f_n)$, and $i = 1$ in Theorem 1.7 (iii). See [Tei77, Exemples d'applications, A)] □

Remark 1.11. It will follow from the proof that the inequalities in Theorem 1.7 hold for $\mathfrak{m}$-primary ideals $\mathfrak{a}_1, \cdots, \mathfrak{a}_n \subset \mathcal{O}$ in any noetherian n-dimensional local ring $(\mathcal{O}, \mathfrak{m})$ having infinite residue field.

Sketch of Proof of Theorem 1.7. We first focus on (i). When $n = 2$ the assertion is that $e(\mathfrak{a};\mathfrak{b})^2 \leq e(\mathfrak{a}) \cdot e(\mathfrak{b})$: this was established by Teissier [Tei73] for normal surfaces and by Rees and Sharp [RS78, Theorem 2.2], for $\mathfrak{m}$-primary ideals in any local noetherian ring $(\mathcal{O}, \mathfrak{m})$ of dimension two. As in the cited papers, we proceed by induction on $n = \dim X$. Specifically, Teissier ([Tei73, p.306]) shows that if $f \in \mathfrak{a}_n$ is sufficiently general, and if one denotes by

$$Y = \mathrm{div}(f) \subset X$$

the hypersurface cut out by f, and $\overline{\mathfrak{a}_i} \subset \mathcal{O}_Y$ the ideals determined by the $\mathfrak{a}_i$, then

$$e_X(\mathfrak{a}_1;\cdots;\mathfrak{a}_n) = e_Y\left(\overline{\mathfrak{a}_1};\cdots;\overline{\mathfrak{a}_{n-1}}\right) \tag{8}$$

where we are using subscripts to indicate the scheme on which the multiplicities are computed. By induction, one finds that

$$\begin{aligned} e_X(\mathfrak{a}_1;\cdots;\mathfrak{a}_n)^{n-1} &= e_Y\left(\overline{\mathfrak{a}_1};\cdots;\overline{\mathfrak{a}_{n-1}}\right)^{n-1} \\ &\leq e_Y\left(\overline{\mathfrak{a}_1}\right)\cdots e_Y\left(\overline{\mathfrak{a}_{n-1}}\right) \\ &= e_X\left(\mathfrak{a}_1^{[n-1]};\mathfrak{a}_n\right)\cdots e_X\left(\mathfrak{a}_{n-1}^{[n-1]};\mathfrak{a}_n\right), \end{aligned}$$

where in the last line we have used (8) again to express the multiplicity of $\overline{\mathfrak{a}_i}$ on Y as a mixed multiplicity on X. The resulting inequality

$$e\left(\mathfrak{a}_1;\cdots;\mathfrak{a}_n\right)^{n-1} \leq e\left(\mathfrak{a}_1^{[n-1]};\mathfrak{a}_n\right)\cdots e\left(\mathfrak{a}_{n-1}^{[n-1]};\mathfrak{a}_n\right)$$

of mixed multiplicities on X is the analogue of equation (2) in the proof of Theorem 1.1, and as in that argument the inequality appearing in (i) now follows formally.

For (ii), if $h_1,\cdots,h_{n-p} \in \mathfrak{b}$ are sufficiently general, and if denote $Y = \mathrm{div}(h_1) \cap \cdots \cap \mathrm{div}(h_{n-p}) \subset X$, then

$$\begin{aligned} e_X\left(\mathfrak{a}^{[q]};\mathfrak{b}^{[n-q]}\right)^p &= e_Y\left(\overline{\mathfrak{a}}^{[q]};\overline{\mathfrak{b}}^{[p-q]}\right)^p \\ &\leq e_Y\left(\overline{\mathfrak{a}}\right)^q \cdot e_Y\left(\overline{\mathfrak{b}}\right)^{p-q} \\ &= e_X\left(\mathfrak{a}^{[p]};\mathfrak{b}^{[n-p]}\right)^q \cdot e_X(\mathfrak{b})^{p-q}. \end{aligned}$$

Statement (iii) is the special case $q = i$ and $p = n$ of (ii).

For (iv), if $h_1,\cdots,h_{i-1} \in \mathfrak{a}$ are sufficiently general, and if denote $Y = \mathrm{div}(h_1)\cap\cdots\cap\mathrm{div}(h_{i-1})$,

then

$$\begin{aligned} m_i^2 &= e_Y\left(\overline{\mathfrak{a}}; \overline{\mathfrak{b}}^{[n-i]}\right)^2 \\ &\leq e_Y\left(\overline{\mathfrak{a}}^{[2]}; \overline{\mathfrak{b}}^{[n-i-1]}\right) \cdot e_Y\left(\overline{\mathfrak{b}}\right) \\ &= m_{i+1} m_{i-1}. \end{aligned}$$

□

2 Amplitude for a Mapping

By way of preparation, consider a proper mapping $f\colon X \to T$ of schemes, and a coherent sheaf $\mathcal{F}$ on X. Then $f_*\mathcal{F}$ is a coherent sheaf on T by [SP26, Tag 02O5], and so one can form the **projective bundle associated to** $f_*\mathcal{F}$

$$\mathbb{P}(f_*\mathcal{F}) \coloneqq \underline{\mathrm{Proj}}_T(\mathrm{Sym}(f_*\mathcal{F})) \longrightarrow T$$

Note that from [SP26, Tag 01O3], for $t \in T$, we have

$$\mathbb{P}(\mathcal{F})_t \cong \mathrm{Proj}\,\mathrm{Sym}\left((f_*\mathcal{F})_t \otimes_{\mathcal{O}_{T,t}} k(t)\right).$$

That is, the fibre over a given point $t \in T$ is the projective space of one-dimensional quotients of the fiber $(f_*\mathcal{F})_t \otimes_{\mathcal{O}_{T,t}} k(t)$. This is the analogue in the relative setting of the projective space of sections of a sheaf on a fixed complete variety. Moreover, there is a natural mapping $f^*f_*\mathcal{F} \to \mathcal{F}$ whose surjectivity is the analogue of the global generation of $\mathcal{F}$ in the absolute situation. Indeed, for $T = \mathrm{Spec}\,k$, we have $f^*f_*\mathcal{F} = f^*H^0(X,\mathcal{F})^\sim = H^0(X,\mathcal{F}) \otimes_k \mathcal{O}_X$, the natural mapping corresponds to the classical evaluation map.

These remarks motivate

Definition 2.1 (Amplitude for a map)**.** Let $f\colon X \to T$ be a proper mapping of algebraic varieties or schemes, and let L be a line bundle on X.

(i). L is **very ample relative to** f, or f**-very ample**, if the canonical map

$$\rho\colon f^*f_*L \longrightarrow L$$

is surjective and defines an embedding

$$\begin{array}{ccc} X & \xhookrightarrow{\ j\ } & \mathbb{P}\left(f_*L\right) \\ & {\scriptstyle f}\searrow \quad \swarrow & \\ & T & \end{array}$$

of schemes over T.

(ii). L is **ample relative to** f, or f**-ample**, if $L^{\otimes m}$ is f-very ample for some $m > 0$.

A Cartier divisor D on X is very ample for f if the corresponding line bundle is so, and f-amplitude for Cartier $\mathbb{Q}$-divisors is defined by clearing denominators.

Example 2.2. If E is a vector bundle on a scheme T, then the Serre line bundle $\mathcal{O}_{\mathbb{P}(E)}(1)$ on $X = \mathbb{P}(E)$ is ample for the natural mapping $\pi\colon \mathbb{P}(E) \to T$.

Proof. We prove $\mathcal{O}_{\mathbb{P}(E)}(1)$ is π-very ample. From [SP26, Tag 01XX], the map $E \to \pi_*\mathcal{O}_{\mathbb{P}(E)}(1)$ is an isomorphism. So $\pi^*\pi_*\mathcal{O}_{\mathbb{P}(E)}(1) \to \mathcal{O}_{\mathbb{P}(E)}(1)$ is identified as $\pi^*E \to \mathcal{O}_{\mathbb{P}(E)}(1)$, which is surjective by [SP26, Tag 01O4]. Note that $\mathbb{P}(E) \to \mathbb{P}(E)$ is the identity, we conclude $\mathcal{O}_{\mathbb{P}(E)}(1)$ is π-very ample, hence π-ample. □

We start by recording several useful observations.

Remark 2.3 (Amplitude is local on the base)**.** Observe that both properties in Definition 2.1 are local on T. In other words, either condition holds for $f\colon X \to T$ if and only if it holds for each of the restrictions $f_i\colon X_i = f^{-1}(U_i) \to U_i$ of f to the inverse images of the members of an open covering $\{U_i\}$ of T.

Remark 2.4 (Equivalent condition for f-very ample)**.** The condition in Definition 2.1 (i) is equivalent to the existence of a coherent sheaf $\mathcal{F}$ on T, plus an embedding $i\colon X \hookrightarrow \mathbb{P}(\mathcal{F})$ over T, such that $L = \mathcal{O}_{\mathbb{P}(\mathcal{F})}(1)\big|_X$. See [GW20, Proposition 13.56].

It follows in particular that if T is affine — so that f_*L is globally generated — then L is very ample for f if and only if there is an embedding $j\colon X \hookrightarrow \mathbb{P}^N \times T$ such that $L = j^*\mathcal{O}_{\mathbb{P}^N \times T}(1)$. This is taken as the definition of very ample relative to a mapping in [Har77].

Remark 2.5. The condition in Definition 2.1 (ii) does not appear as the definition of f-amplitude in [DG61], but it is equivalent to that definition by virtue of [DG61, II, 4.6.11].

The cohomological properties of relative amplitude are the following:

Theorem 2.6. Let $f\colon X \to T$ be a proper morphism of schemes, and L a line bundle on X. Then the following are equivalent:

(i). L is ample for f.

(ii). Given any coherent sheaf $\mathcal{F}$ on X, there exists a positive integer $m_1 = m_1(\mathcal{F})$ such that

$$R^i f_* \left(\mathcal{F} \otimes L^{\otimes m}\right) = 0 \text{ for all } i > 0,\ m \geq m_1(\mathcal{F}).$$

(iii). Given any coherent sheaf $\mathcal{F}$ on X, there is a positive integer $m_2 = m_2(\mathcal{F})$ such that the canonical mapping

$$f^* f_* \left(\mathcal{F} \otimes L^{\otimes m} \right) \longrightarrow \mathcal{F} \otimes L^{\otimes m}$$

is surjective whenever $m \geq m_2$.

(iv). There is a positive integer $m_3 > 0$ such that $L^{\otimes m}$ is f-very ample for every $m \geq m_3$.

Example 2.7. Suppose given a diagram

$$\begin{array}{ccc} Y & \xrightarrow{\quad\mu\quad} & X \\ & {\scriptstyle g}\searrow \quad \swarrow{\scriptstyle f} & \\ & T & \end{array}$$

of schemes over T, with μ being finite. If L is an f-ample line bundle on X, then $\mu^* L$ is ample relative to g.

Proof. Since L is f-ample, then there is an open covering $\{U_i\}$ of T such that $L|_{f^{-1}(U_i)}$ is an ample line bundle in the absolute sense by [GW20, Theorem 13.62 and Proposition 13.63]. But as the restriction

$$\mu|_{g^{-1}(U_i)} : \mu^{-1}\left(f^{-1}(U_i)\right) = g^{-1}(U_i) \longrightarrow f^{-1}(U_i)$$

is finite, so by finite pullbacks I we see $\mu^* L|_{g^{-1}(U_i)}$ is absolutely ample. We see $\mu^* L$ is g-ample. □

The following useful result allows one in practice to reduce to the absolute setting:

Theorem 2.8 (Fibre-wise amplitude)**.** Let $f\colon X \to T$ be a proper morphism of schemes, let L be a line bundle on X, and for $t \in T$ set

$$X_t = f^{-1}(t), \quad L_t = L|_{X_t}.$$

Then L is ample for f if and only if L_t is ample on X_t for every $t \in T$.

Proof. First assume L is f-ample. There is a commutative diagram

$$\begin{array}{ccc} X_t & \xhookrightarrow{\quad\mu\quad} & X \\ & {\scriptstyle g}\searrow \quad \swarrow{\scriptstyle f} & \\ & T & \end{array}$$

where μ is a closed immersion, hence finite. By Example 2.7, $\mu^* L = L|_{X_t}$ is g-ample. But g sends X_t to a single closed point, so this relative amplitude is nothing but the absolute meaning.

Conversely, suppose that L_t is ample for every $t \in T$. By the proof of amplitude in families, every point $t \in T$ has a neighborhood U_t with the property that

$$L^{\otimes m_t}\big|_{f^{-1}(U_t)} \cong \phi_{U_t}^* \left(\mathcal{O}_{\mathbb{P}^r \times U_t}(1)\right)$$

for a finite morphism $X_{U_t} \coloneqq f^{-1}(U_t) \xrightarrow{\phi_{U_t}} \mathbb{P}^r \times U_t$ of schemes over U_t. It follows from Example 2.7, that for

$$\begin{array}{ccc} X_{U_t} & \xrightarrow{\phi_{U_t}} & \mathbb{P}^r \times U_t \\ & {\scriptstyle f_{U_t}} \searrow \quad \swarrow {\scriptstyle \pi} & \\ & U_t & \end{array}$$

with $\mathcal{O}_{\mathbb{P}^r \times U_t}(1)$ is π-ample, so $L|_{f^{-1}(U_t)}$ is f_{U_t}-ample on each U_t of $t \in T$. Recall that f-amplitude is local on T, we see L is indeed ample with respect to f. □

Corollary 2.9 (Nakai's criterion for a mapping)**.** In the setting of the Theorem 2.8, a $\mathbb{Q}$-divisor D on X is ample with respect to f if and only if $\left(D^{\dim V} \cdot V\right) > 0$ for every irreducible subvariety $V \subset X$ of positive dimension that maps to a point in T.

The next result summarizes the connection between relative and global amplitude.

Proposition 2.10. Consider a morphism

$$f \colon X \longrightarrow T$$

of projective schemes. Let L be a line bundle on X, and let A be an ample line bundle on T. Then L is f-ample if and only if $L \otimes f^* \left(A^{\otimes m}\right)$ is an ample line bundle on X for all $m \gg 0$.

Proof. Assume that L is f-ample. Note that $L \otimes f^* \left(A^{\otimes m}\right)$ is ample if and only if its higher power is ample, up to replacing L by its higher power, we can suppose that $f^* f_* L \to L$ is surjective. Since A is ample, $f_* L \otimes A^{\otimes p}$ is globally generated for p sufficiently large by [SP26, Tag 01Q3]. Therefore its pullback $f^* f_* L \otimes f^* A^{\otimes p}$ is likewise globally generated since f^* is right exact (on evaluation map). So we choose generators give rise to a morphism

$$\phi \colon X \longrightarrow \mathbb{P} \times T$$

of schemes over T, where $\mathbb{P} = \mathbb{P} H^0 \left(T, f_* L \otimes A^{\otimes p}\right)$ and ϕ is induced from the universal property of the fiber product. Note that we have $L \otimes f^* A^{\otimes p} \cong \phi^* \operatorname{pr}_1^* \mathcal{O}_{\mathbb{P}}(1)$. For arbitrary $t \in T$, ϕ restrict on X_t is

$$\phi_t \coloneqq \phi|_{X_t} : X_t \longrightarrow \mathbb{P} \times \{t\} \cong \mathbb{P}.$$

Then $L_t \cong L_t \otimes f_t^* \left(A_t^{\otimes p}\right) \cong \left(L \otimes f^* \left(A^{\otimes p}\right)\right)_t \cong \phi_t^* \mathcal{O}_{\mathbb{P}}(1)$ says that ϕ_t is fiber finite, since if $C \subset X_t$ is a curve contracted by ϕ_t, then

$$L_t|_C \cong \phi_t^* \mathcal{O}_{\mathbb{P}}(1)|_C = \left(\phi_t|_C\right)^* \mathcal{O}_{\mathbb{P}}(1) \cong \mathcal{O}_C,$$

contradicting to L_t ample (Theorem 2.8). Combine with ϕ proper, ϕ is finite. Therefore

$$L \otimes f^* \left(A^{\otimes (p+1)}\right) = \phi^* \left(\mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}}(1) \otimes \mathrm{pr}_2^* A\right)$$

is the pullback of an ample line bundle under a finite map, and consequently is ample.

The converse follows from Theorem 2.8, since we have

$$L_t \cong L_t \otimes f_t^* \left(A_t^{\otimes m}\right) \cong \left(L \otimes f^* \left(A^{\otimes m}\right)\right)_t$$

where f pulls back a trivial line bundle. □

Finally, we say a few words about nefness for a mapping. Here one takes the analogue of Theorem 2.8 as the definition:

Definition 2.11 (Nefness relative to a mapping)**.** Given a proper morphism $f\colon X \to T$ as above, a line bundle L on X is **nef relative to** f if the restriction $L_t = L|_{X_t}$ of L to each fiber is nef, or equivalently if $(c_1(L) \cdot C) \geq 0$ for every $C \subset X$ mapping to a point under f.

Note that the analogue of Proposition 2.10 need not be true.

Example 2.12. Let $X = E \times E$ be the product of an elliptic curve with itself and $f\colon X \to E$ the first projection. Let $D = E \times \{\mathrm{pt}\}$ and denote by $\Delta \subset E \times E$ the diagonal. Then $D - \Delta$ is f-nef. However, for any divisor A on E the divisor

$$D - \Delta + f^*(A)$$

has self-intersection -2, and hence cannot be nef.

Remark 2.13 (Fujita vanishing for a mapping)**.** In his paper [Kee03], Keeler extends Fujita's Vanishing Theorem 1.4.35 to the relative setting.

Remark 2.14 (Other ground fields)**.** Everything in this section goes through to varieties defined over an algebraically closed field of arbitrary characteristic.

References

[DG61] Jean Dieudonné and Alexander Grothendieck, *Éléments de géométrie algébrique II: Étude globale élémentaire de quelques classes de morphismes*, *Publications Mathématiques de l'IHÉS* 8 (1961), 5–222.

[Ful98] William Fulton, *Intersection Theory*, 2nd ed., Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, Vol. 2, Springer-Verlag, New York, 1998.

[GW20] Ulrich Görtz and Torsten Wedhorn, *Algebraic Geometry I: Schemes. With Examples and Exercises*, 2nd ed., Springer Studium Mathematik – Master, Springer Spektrum, Wiesbaden, 2020.

[Har77] Robin Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics, Vol. 52, Springer-Verlag, New York, 1977.

[Kee03] Dennis S. Keeler, Ample filters of invertible sheaves, *Journal of Algebra* 259 (2003), no. 1, 243–283.

[Laz04] Robert Lazarsfeld, *Positivity in Algebraic Geometry I: Classical Setting: Line Bundles and Linear Series*, Ergebnisse der Mathematik und ihrer Grenzgebiete, Vol. 48, Springer-Verlag, Berlin–Heidelberg, 2004.

[RS78] David Rees and Rodney Y. Sharp, On a theorem of B. Teissier on multiplicities of ideals in local rings, *Journal of the London Mathematical Society* (2) 18 (1978), no. 3, 449–463. https://doi.org/10.1112/jlms/s2-18.3.449

[SP26] The Stacks Project Authors, *The Stacks Project*, 2026, https://stacks.math.columbia.edu/.

[Tei73] Bernard Teissier, Cycles évanescents, sections planes et conditions de Whitney, in *Singularités à Cargèse*, Astérisque, nos. 7–8, 1973, 285–362. https://www.numdam.org/item/AST_1973__7-8__285_0/

[Tei77] Bernard Teissier, Sur une inégalité à la Minkowski pour les multiplicités, Appendix to David Eisenbud and Harold I. Levine, An algebraic formula for the degree of a C^∞ map germ, *Annals of Mathematics* (2) 106 (1977), no. 1, 38–44. https://doi.org/10.2307/1971156